Skills and Competences for the Sustainability Engineer

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Abstract

Today's engineering education is pressurised to respond to the Sustainable Development Goals (SDGs), yet the literature lacks consensus on which competences future engineers actually need. This short paper analyses[1] European focus-group study—covering 86 students, academics, and employers across Denmark, Finland, France, and Ireland—to examine three competences relevant to a sustainability engineer:**systems thinking**, **life cycle thinking**, and **interpersonal competence**. Upon analysis, the paper shows strong stakeholder agreement that systems thinking and interpersonal competence are essential, while life cycle thinking is recognised primarily by students and remains implicit. The analysis questions the notion of a "typical" sustainability engineer, arguing that no single profile exists; instead, the typical engineer is a technically grounded professional who adds **holistic** and **human-centred** competences to disciplinary depth. The analysis concludes that a gap is not in identifying competences but in integrating them into overcrowded curricula.

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1 Introduction

Engineering problems are described in the literature as "wicked" problems—complex, multi-faceted, and resistant to purely technical solutions[1]. Future Engineers are expected to act as catalyst of technical innovation and lead on socioeconomic issues. The UN Sustainable Development Goals (SDGs) provide a framework for this shift, covering 169 targets across environmental, social, and economic domains. However, no agreement on which competences engineering graduate should develop to meet these challenges.

This paper analyses selected literature on the skills and competences required by engineers for sustainability, using **Beagon at al. (2023)** as primary source. Three competences for the typical engineer for sustainability is addressed. The paper first outlines the methodological approach, then analyses systems thinking, life cycle thinking, and interpersonal competence, before discussing the concept of the "typical" engineer.

2 Methodology

The study employed a qualitative design using twelve focus groups conducted in Denmark, Finland, France, and Ireland between March and July 2019, involving 86 participants from three stakeholder groups: engineering academics, employers, and students. Data was analysed through an open coding process, producing 54 competences grouped into six categories: Fundamental Technical Skills, Application Skill, Outward facing People-Oriented Skills, Inward-facing Ways of thinking, World view, and Character and Ethical Orientation and mapped against the UNESCO (2017) key competences framework.

For the this paper, three competences were selected from the study's findings to represent different sustainability practice: system thinking (a way of thinking), life cycle thinking (an application skill), and interpersonal competence (a people-oriented skill).

3 Analysis of Three Key Competences

3.1 System Thinking

It appears in the findings under the category of "Inward-Facing Ways of Thinking," operationalised as analytical and holistic thinking. Holistic thinking is defined by participants as "the capacity to make a good synthesis and see the whole picture." All three stakeholder groups—students, academics, and employers—identified systems thinking as a key competence.

Its relevance to sustainability is grounded in the nature of sustainability problems themselves. Because engineering challenges are "ill-defined, open-ended, cross-disciplinary, complex socio-technical problems," an engineer who optimises a single component in isolation risks shifting burden elsewhere in the system One academic quoted in the study described the shift required: students in university "do what we say to do" without a risk or responsibility, but in professional practice "they have a responsibility and they have to be able to justify their actions and apply global thinking". Another stakeholder emphasised the ability to "perceive the worldwide situation, not only from the engineer's perspective, but also from other viewpoints."

System thinking therefore functions as the cognitive foundation of sustainability engineering: it is the competence that allows an engineer to situate a technical decision within environmental, social, and economic systems simultaneously.

3.2 Life Cycle Thinking

This is the ability to consider a product, process, or infrastructure across its entire life—from material extraction through use to disposal.

Its relevance is closely tied to what the study identifies as a notable gap: **anticipatory competence**. Beagon et al.found that none of the stakeholder groups explicitly identified anticipatory competence—the ability to envision future scenarios and estimate consequences—despite it begin one of UNESCO' (2017) eight key competences. The authors argue that this competence is "not entirely absent" because similar ideas appear implicitly in other competences, specifically naming life cycle thinking, lifelong learning, and time management as carriers of future-oriented thinking.

A striking finding is the divergence between stakeholder groups. Life cycle thinking was identified by **students** as a necessary competence, but not explicitly by **academics** or **employers**. This suggests that students—who will inherit the long-term consequence of today's design decisions—may already be internalising the future-oriented perspective that the literature consider essential, even where industry and curricular have not yet formalised it.

3.3 Interpersonal Competence

This competence mainly encompasses communication, collaboration, teamwork, and related abilities such as listening, emotional intelligence, and empathy. The study describes interpersonal competence as "competences associated with engaging with people, on a general level, regardless of their discipline".

This is arguably the study's most emphatic finding. Its relevance to sustainability is practical: sustainability solutions require negotiation across diverse value sets. As one employer explained, "engineers will also need to manage divers project teams, there are no longer only engineers on teams[..] They now include environmentalists, economists, institutional capacity builders, all kinds of people."

Yet the study also documents a persistent gap. One employer stated that "some of them come in lacking in those skills, particularly compared to business graduates."Academics noted that while collaboration is taught implicitly through project work, competences such as empathy and emotional intelligence are "much needed but not sufficiently promoted in engineering curricula." The study concludes that there needs to be "more emphasis and explicit assessment of interpersonal competences ... rather than relying on gaining these competences implicitly through project work."

4 Discussion

The term "typical" is difficult to apply to the sustainability engineer, and the literature suggests it may be misleading if understood as a distinct job title. Beagon et al.'s findings support three observations.

First, technical depth remains non-negotiable. One academic stated that "hard-core, solid technical skills are required—however, if engineers have technical skills only that is clearly inadequate," while an employer warned that universities must not compromise technical training. The "typical" sustainability engineer is therefore not a generalist who has traded depth for breadth; the study explicitly rejects this, noting that "not all people can be generalists" and that each engineer "has to understand how his/her special expertise is connected to others."

Second, the competences identified are additive rather than substitutive. Systems thinking, life cycle thinking, and interpersonal competence are layered on top of disciplinary expertise. This aligns with the T-shaped profile: depth in one discipline (the vertical bar) combined with the ability to connect across disciplines and stakeholders (the horizontal bar). The "typical" sustainability engineer is thus a mechanical, civil, or electrical engineer who applies sustainability competences within their domain—not a separate profession.

Third, the study reveals that the "typical" profile is still contested. Employers called for mathematics and interpretation skills and for graduates who "challenge the status quo"; academics emphasised conceptual understanding and negotiation; students valued organisation, leadership, and life cycle thinking that neither of the other groups named. The absence of anticipatory competence from all groups further complicates the picture, since a future-oriented competence is precisely what sustainability demands.

The most defensible definition, therefore, is that the "typical" sustainability engineer is defined less by a fixed role than by a mindset: a technically competent professional who refuses siloed thinking, considers long-term consequences, and can work with people who do not share their disciplinary background.

5 Conclusion

This paper analysed Beagon et al.'s (2023) study of engineering competences for the SDGs and discussed three relevant competences. Systems thinking provides the cognitive framework for complex socio-technical problems, life cycle thinking extends that framework across time, and interpersonal competence enables engineers to negotiate sustainability solutions with diverse stakeholders. The "typical" sustainability engineer is therefore not a separate profession, but a technically grounded engineer who adds holistic, future-oriented, and human-centred competences to disciplinary depth[1]. The real barrier is not identifying these competences, but integrating them into already overcrowded curricula.

References

[1] U. Beagon et al. “Preparing engineering students for the challenges of the SDGs: what competences are required?” In: *European Journal of Engineering Education* 48.1 (2023), pp. 1–23.